CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VIth Semester of B.Sc. Biochemistry Examination April-May 2019

BC604: Metabolism II

Date: 27/04/2019 Day: Saturday Time: 1:30 pm To 1:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		10
1.	In amino acid catabolism, the first reaction for many amino acids is a__________.		
	(i) decarboxylation	(ii) transamination	
	(iii) hydroxylation	(iv) Deamination	
2.	Which substance is not involved in the production of urea from $NH4^+$ via the urea cycle?		
	(i) Aspartate	(ii) ATP	
	(iii) Ornithine	(iv) Carbomoyl phosphate	
3.	Phenylalanine is synthesized from_________.		
	(i) 4-Hydroxyphenylpyruvate	(ii) Phenylpyruvate	
	(iii) both i and ii	(iv) none of the above	
4.	Amino acids involved in biosynthesis of glutathione are ______________.		
	(i) Glutamine, Glycine, Glutamate	(ii) Glutamine, Cysteine, Glutamate	
	(iii) Glutamate, Cysteine, Glycine	(iv) Cysteine, Glycine, Lysine	
5.	A balance maintained between nitrogen fixation and atmospheric nitrogen by bacteria under anaerobic condition is called denitrification. (True/False)		

6.	Dying erythrocytes in spleen is degraded to yield___________.		
	(i) Fe^{3+}	(ii) Fe^{4+}	
	(iii) Fe^{+}	(iv) Fe^{2+}	
7.	Which amino acid contributes the amino group for purine and pyrimidine both?		
	(i) Glycine	(ii) Glutamate	
	(iii) Aspartate	(iv) Glutamine	
8.	All aminotransferases have the different prosthetic group and same reaction mechanism. (True/False)		
9.	Which of the following is an analogue of hypoxynthine?		
	(i) PRPP	(ii) Arabinose	
	(iii) Allopurinol	(iv) oxypurinol	
10.	Synthesis of glutamine usually occur in_________.		
	(i) Liver	(ii) Brain	
	(iii) Kidney	(iv) All of the above	

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VIth Semester of BSc Biochemistry Examination April-May 2019

BC604: Metabolism II

Date: 27/04/2019 Day: Saturday Time: 1:50 pm To 3:30 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in TWO SEPARATE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I		
Q - II Answer the following questions as directed.		**10**
Q-IIA	**Answer the following questions as directed [Any Two].**	**6**
1	Explain the catabolic pathway of alanine, glycine, serine and threonine.	
2	Explain the nitrogen fixation and electron path in nitrogenase complex.	
3	Discuss the biosynthesis of neurotransmitters.	
Q-IIB	**Answer the following questions as directed [Any one].**	**4**
1	Describe the Krebs bicycle for ammonia excretion.	
2	Discuss the catabolic pathway of pyrimidine.	
SECTION - II		
Q-III Answer the following questions [Any Three].		**15**
1	Biosynthesis pathway of purine nucleotide.	
2	Explain the integrated pathway of amino acid metabolism.	
3	Write a note on ribonucleotide reductase.	
4	Describe the biosynthesis of Methionine.	